

Student Guide — primary / module-09-subtraction-nikhilam

IKS Curriculum

Table of Contents

Student Workbook

Student Workbook — Subtraction — Nikhilam (All from 9) (Primary)

This workbook is yours to write in. Use it for vocab, exercises, and reflection across all 10 days.

Vocabulary tracker

Fill in as we go:

- *Nikhilam Navataścaramam Daśataḥ* — _____
- *pratiyogi (modern coinage)* — _____

Day-by-day notes

Day 1 — The Sūtra — All from 9, Last from 10

What I learned:

One question I have:

Day 2 — Complement to 1000 and Beyond

What I learned:

One question I have:

Day 3 — Subtract from a Power of 10

What I learned:

One question I have:

Day 4 — General Subtraction via Complements

What I learned:

One question I have:

Day 5 — Why It Works — The Algebra

What I learned:

One question I have:

Day 6 — Bar Model Proof

What I learned:

One question I have:

Day 7 — Two's Complement in Computers

What I learned:

One question I have:

Day 8 — Speed Day

What I learned:

One question I have:

Day 9 — Mixed Practice

What I learned:

One question I have:

Day 10 — Final Assessment

What I learned:

One question I have:

Final reflection

After Day 10, write a paragraph: what is the most important thing you learned in this module?

Day 1 — Friends to Ten

Module: Friends to Ten (Primary) · **Time:** 35 min

Goal

Children meet “friends to ten” through a chant, a story, and a card game. By end of class they can name at least three friend-pairs.

Materials

- A deck of number cards 0–9 (4 of each = 40 cards) per pair
- Drawing paper + crayons
- “Friends to Ten” wall poster (5 colourful pairs)

The flow

Settle + chant (5 min)

Sit on the mat. Teacher: > *“For the next two weeks we’ll meet some friends. They’re tiny — but very important. Numbers have friends too.”*

Teach the chant (3 times, claps on each pair):

“1 and 9 are friends! 2 and 8 are friends! 3 and 7 are friends! 4 and 6 are friends! 5 and 5 are best friends! All make TEN!”

Story (5 min)

The Friend Game

Ten little ants lived on a leaf. Sometimes they walked off in groups.

One day, 3 ants walked off. (*Hold up 3 fingers.*) How many were left behind on the leaf? (*Wait. Children call out: 7!*)

Yes! 3 and 7 are friends. They always make 10 together.

Another day, 6 ants walked off. (*Hold up 6.*) How many on the leaf? (*4!*)

6 and 4 are friends. Always 10.

Every number from 0 to 10 has a friend. We’ll meet them all today.

Card game: Make Ten (20 min)

How to play (paired):

1. Shuffle. Deal 5 cards to each player. Put 5 face-up in the middle.
2. On your turn:
 - Look at your hand and the middle.
 - Find two cards that are friends (add to 10).
 - Take both. Put in your pile.

- Draw 1 from the deck.
- 3. If you can't make 10, place 1 card in the middle. End turn.
- 4. When the deck runs out, count pairs. Most pairs wins.

Play 2–3 rounds.

Show + chant (5 min)

- “Who won a round?” — winner shares ONE friend pair they made.
- Class chants the pair: “6 and 4 are friends!”
- Close with the daily chant.

Homework / take-home

- Practise the chant with a sibling or parent. Win a round of Make Ten with family!

Differentiation

- **Tier up:** Use 0–9 cards including 0+10 pairs.
- **Tier down:** Use cards 1–6 to keep pairs simple.

Teacher's quick notes

- Some children won't find pairs and will give up. Pair them with a stronger player as a “team.”
- Same game as Module 5 — children may come in already half-knowing it. Affirm and add the *friend* framing.
- The “friend” language is the bridge to subtraction. By Day 4 we'll say “to find 100 minus 7, ask: who's 7's friend? 3. Then put a 9 in front. 93.”

Teacher note on Sanskrit

We don't introduce *Nikhilam* yet. The word appears on Day 3 as a “magic word.” Today is just about friends.

Day 2 — Hugs to Ten

Module: Friends to Ten (Primary) · **Time:** 35 min

Goal

Children practise friend recall in a paired drill until every child can name a friend in under 2 seconds.

Materials

- Number cards 0–9 (one set per pair)
- Stopwatch (optional)

The flow

Settle + chant (5 min)

The friends-to-ten chant, 3 times.

Story-warm-up (3 min)

“Yesterday we met the friends. Today they want to HUG. When two friends meet, they hug and become TEN.”

Demonstrate: hold up a “3” card. Pause. “*Who’s 3’s friend?*” Children chorus: SEVEN! Bring out a “7” card. Hug them together. “*Now they’re TEN.*”

Paired drill (20 min)

Round 1 — Quick recall (8 min): - One child holds the deck. Flips card. - Other child names the friend. - Stack hits / misses in two piles. - Swap roles after 30 seconds. - Goal: 10 right in a row.

Round 2 — Speed (5 min): - Time how many friends you can name in 30 seconds. - Both children try.

Round 3 — Hide the friend (7 min): - Lay 5 cards face-up. Hide one (turn over). - Partner has to GUESS the hidden card based only on its friend. - E.g., visible cards are 1, 3, 5, 7, 9; sum should be 25. Actual sum visible = ? Difference tells you the hidden card. - (For Grade 3, simplify: just show one card’s friend; partner names what’s hidden.)

Wrap-up (7 min)

- Quick chorus: teacher calls a digit; class chorus the friend. Do 15 in 60 seconds.
- 4 → SIX! 8 → TWO! 1 → NINE! 6 → FOUR! 2 → EIGHT! 5 → FIVE! ...
- Daily chant.

Homework / take-home

- Friend drill with family: 20 digits in 60 seconds. Track how many you got.

Differentiation

- **Tier up:** “Friends to 100” preview: who completes 73? (27.) Show the friend on bar paper.
- **Tier down:** Use only 1–5 cards. Friends will be larger but the count is shorter.

Teacher’s quick notes

- The HUG metaphor is sticky for this age group. Hold it.
- Some children laugh at “5 and 5 are best friends.” Let them. The laugh helps the math stick.
- Avoid timed competition between pairs publicly. Each pair races itself.

Day 3 — The Magic Word

Module: Friends to Ten (Primary) · **Time:** 35 min

Goal

Children meet the Sanskrit word *Nikhilam*. They learn to pronounce it, recognise the wall poster, and connect it to “friends to ten.”

Materials

- A new wall poster: the word **Nikhilam** in big letters, with a smiling face. Underneath, in small letters: “the friend rule.”
- Drawing paper
- Crayons

The flow

Settle + chant (5 min)

The friends-to-ten chant, 3 times.

The magic-word story (10 min)

The Sanskrit Word

Long ago, people in India had a special name for “the friend rule.” The name is a Sanskrit word.

(Pause for suspense.)

Are you ready?

The word is — **Nikhilam!** *(Say it slowly: ni-khi-lam.)*

(Class echoes 3 times. Soft / normal / loud. Children find it funny — let them.)

Nikhilam means “all of it” or “the whole.” It’s the secret name for our friends-to-ten rule.

An older book from 1965 gives this rule a fancy name. We’ll learn the long name in Grade 6. For today — just say *Nikhilam*. The friend rule. The whole. All of it.

Drawing time (15 min)

Each child draws ONE friend-pair. Above the pair, write the word **NIKHILAM** in big letters (or use the wall poster as a guide).

Examples: a pair of trees labelled “3 and 7.” Two children holding hands labelled “4 and 6.” Two suns labelled “5 and 5.”

(Teacher walks around helping with the spelling of “Nikhilam.”)

Sharing (5 min)

3 children hold up their drawings. Class chorus the friend-pair AND chants *Nikhilam* together.

End with the daily chant + one big “*NIKHILAM!*”

Homework / take-home

- Tell a parent the magic word: *Nikhilam*. Say it 5 times. Win their amazement.

Differentiation

- **Tier up:** Try to write *Nikhilam* in Devanāgarī (निखिलं). Provide a print-out card.
- **Tier down:** Just say the word. Drawing is optional.

Teacher’s quick notes

- The pronunciation is harder than it looks. Model slowly. Don’t correct individual children publicly — they’ll get it by Day 6 through repetition.
- Some children will ask “what does it mean?” Answer: “all” or “the whole” or “every part.” The full meaning becomes clear in Middle band.
- This is the first Sanskrit word many children will say with confidence. Honour it. Some will go home and say it 200 times.

Citation(s) used (teacher reference, not student-facing)

- Tirthaji, *Vedic Mathematics* (1965), sūtra Nikhilam.

Day 4 — 100 minus Friends

Module: Friends to Ten (Primary) · **Time:** 35 min

Goal

Children compute $100 - N$ for single-digit N using “friends to ten” and a bar model. By end of class: $100 - 7 = 93$, instantly.

Materials

- Bar-model strips: long printed strips divided into 10 boxes of 10 each (= 100 squares total)
- Coloured crayons
- Number cards 0–9

The flow

Settle + chant (5 min)

The friends-to-ten chant. A new line: “*And TEN tens make ONE HUNDRED!*”

Bar-model demo (10 min)

Draw a big bar on the board (or use the strip). Mark off 10 sections of 10 each. Total: 100.

Teacher: “100 ants on a hundred leaves. 7 ants walk off. How many left?”

Cross off 7 squares from the rightmost group. “This last group used to be 10. Now it’s 3.”

“How many groups of 10 still left? NINE groups. Plus the 3 leftover ants. So $90 + 3 = 93$.”

The trick: 7’s friend is 3 (10 minus 7 = 3). And there are 9 groups of 10 still standing.

So $100 - 7 = 93$.

Demo 3 more, with class predicting:

Problem	Friend of N	9 in front	Answer
$100 - 4$	6	9	96
$100 - 8$	2	9	92
$100 - 1$	9	9	99
$100 - 9$	1	9	91

Activity: bar-model worksheet (15 min)

Each child gets a printed worksheet with 6 bar-model problems:

$100 - 3 = ?$
 $100 - 5 = ?$
 $100 - 6 = ?$
 $100 - 2 = ?$
 $100 - 8 = ?$
 $100 - 7 = ?$

For each: shade the right number of squares, then read off the answer. Bonus: write the friend pair next to it.

Wrap-up (5 min)

- Quick fire: teacher calls $100 - N$; children chorus the answer.
- Daily chant.

Homework / take-home

- 10 problems: $100 - 4$, $100 - 9$, $100 - 6$, $100 - 2$, $100 - 5$, $100 - 8$, $100 - 3$, $100 - 7$, $100 - 1$, $100 - 0$. Aim for under 30 seconds.

Differentiation

- **Tier up:** Try $100 - 10$, $100 - 15$, $100 - 20$. Bar model still works.
- **Tier down:** Use only N from 1 to 5. Build up.

Teacher's quick notes

- The “9 in front” framing is the key insight. Make sure every child says it.
- For $N = 0$ or $N = 10$, the bar model is a bit off (no shading, or a whole group gone). Acknowledge and move on; those are edge cases.
- Some children will sniff out the pattern: “to do 100 minus any single digit, just write ‘9’ then the friend of that digit.” Praise — that IS the Nikhilam rule for $100 - N$.

Day 5 — Why It Works — The Algebra

Module: Subtraction — Nikhilam (All from 9) (Primary) · **Time:** 30 min

Goal

Derive the place-value identity that makes the rule work.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *Nikhilam Navataścaramam Daśataḥ* (IAST: Nikhilam Navataścaramam Daśataḥ; lit. “all from 9, the last from 10”)
- Why does the trick work? Let's see!

Write $100 = 99 + 1$. - $99 - 37$ is easy (no borrow): 6 from 9 = 6 and $9 - 3 = 6$, write it out as: $9 - 3 = 6$, $9 - 7 = 2$... actually let's keep it simple. - $100 - 37 = 63$. - And $9 - 3 = 6$, $10 - 7 = 3 \rightarrow 63$.

The 9s help because there's no borrow. “*That's the magic!*”

Activity (8 min)

A simplified version of *Complement Flashcards* (see activities/). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 6 — Bar Model Proof

Module: Subtraction — Nikhilam (All from 9) (Primary) · **Time:** 30 min

Goal

Draw a visual proof using bar models and number lines.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *pratiyogi* (modern coinage) (IAST: pratiyogi (modern coinage); lit. “the number that adds to a base”)
- Let's see the trick with bars (or paper strips)!

Cut a strip 100 cm long. Now cut off 37 cm. “*How long is the rest?*” (63 cm.)

Now cut a strip 99 cm. Cut off 37 cm. “*How long?*” (62 cm — easy! $9-3=6$, $9-7=2$.) Add 1 cm — 63 cm.

“*Same answer! Easier with 99!*”

Activity (8 min)

A simplified version of *Complement Flashcards* (see activities/). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 7 — Two's Complement in Computers

Module: Subtraction — Nikhilam (All from 9) (Primary) · **Time:** 30 min

Goal

Show how the same idea underlies binary computer subtraction.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: *“What's one thing you remember from yesterday?”*

Today's idea (10 min)

- *Nikhilam Navataścaramaṁ Daśataḥ* (IAST: Nikhilam Navataścaramaṁ Daśataḥ; lit. “all from 9, the last from 10”)
- Computers use the same trick! In computers, numbers are 0s and 1s. They use “all from 1, last from 1+1” — same idea but with 1 instead of 9.

Show a calculator. *“When you press 100 – 37, the computer secretly uses our trick to find the answer fast.”*

Children share: *“What's the coolest computer thing they know?”*

Activity (8 min)

A simplified version of *Complement Flashcards* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 8 — Speed Day

Module: Subtraction — Nikhilam (All from 9) (Primary) · **Time:** 30 min

Goal

Complete a 30-problem mental subtraction sprint.

Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What’s one thing you remember from yesterday?*”

Today’s idea (10 min)

- *pratiyogi* (modern coinage) (IAST: pratiyogi (modern coinage); lit. “the number that adds to a base”)
- Sprint day! 10 problems in 3 minutes.

Each child gets a worksheet: 1. $10 - 4$ 2. $100 - 37$ 3. $100 - 62$ 4. $1000 - 250$ 5. $100 - 88$...

Cheer for everyone who finishes. “*Speed isn’t everything — accuracy matters more!*”

Activity (8 min)

A simplified version of *Complement Flashcards* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 9 — Mixed Practice

Module: Subtraction — Nikhilam (All from 9) (Primary) · **Time:** 30 min

Goal

Solve a mixed problem set comparing complement vs traditional borrow.

Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What’s one thing you remember from yesterday?*”

Today’s idea (10 min)

- *Nikhilam Navataścaramam Daśataḥ* (IAST: Nikhilam Navataścaramam Daśataḥ; lit. “all from 9, the last from 10”)
- Today: pick the right tool!

Show 10 problems on cards. For each, children hold up a sign: - “Trick!” (for subtracting from 10, 100, 1000) - “Regular!” (for harder cases)

Then solve 5 trick-type problems together.

Activity (8 min)

A simplified version of *Complement Flashcards* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 10 — Final Assessment

Module: Subtraction — Nikhilam (All from 9) (Primary) · **Time:** 30 min

Goal

Take the summative quiz.

Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What’s one thing you remember from yesterday?*”

Today’s idea (10 min)

- *pratiyogi* (*modern coinage*) (IAST: pratiyogi (modern coinage); lit. “the number that adds to a base”)
- Final day! We celebrate the subtraction trick.

Each child shows one favourite trick problem to a partner. Cheer for each.

Take the small quiz with stickers. Every child gets a “Subtraction Sleuth” sticker.

End: chant — “*All from nine, last from ten!*”

Activity (8 min)

A simplified version of *Complement Flashcards* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Homework

Homework — Subtraction — Nikhilam (All from 9) (Primary)

Light daily tasks. Each takes 5–10 minutes.

1. **Day 1 — The Sūtra — All from 9, Last from 10** · Draw about today’s concept. (~5 min)
2. **Day 2 — Complement to 1000 and Beyond** · Draw about today’s concept. (~5 min)
3. **Day 3 — Subtract from a Power of 10** · Draw about today’s concept. (~5 min)
4. **Day 4 — General Subtraction via Complements** · Draw about today’s concept. (~5 min)
5. **Day 5 — Why It Works — The Algebra** · Draw about today’s concept. (~5 min)
6. **Day 6 — Bar Model Proof** · Draw about today’s concept. (~5 min)
7. **Day 7 — Two’s Complement in Computers** · Draw about today’s concept. (~5 min)
8. **Day 8 — Speed Day** · Draw about today’s concept. (~5 min)
9. **Day 9 — Mixed Practice** · Draw about today’s concept. (~5 min)
10. **Day 10 — Final Assessment** · Draw about today’s concept. (~5 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.

- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.